Chiziqli ifodalarda qavslarni ochish

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Soddalashtiring:** $-3(2x-5)+4(x-2)$

A) $-2x+7$ **B)** $-2x-23$ **C)** $-2x+13$ **D)** $10x+7$

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2. **Soddalashtiring:** $5(a-2b)-2(3a-4b)$

A) $-a-18b$ **B)** $-a+2b$ **C)** $-a-2b$ **D)** $11a-2b$

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3. **Soddalashtiring:** $x(2y-3)-3x(y-1)$

A) $-xy+6x$ **B)** $-xy$ **C)** $-xy-6x$ **D)** $5xy-6x$

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4. **Soddalashtiring:** $3[2x-5(x-2)]$

A) $-9x+10$ **B)** $-9x-30$ **C)** $-9x+30$ **D)** $21x-30$

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5. **Soddalashtiring:** $6m-[2m-3(m+2)]$

A) $7m+6$ **B)** $7m-6$ **C)** $m-6$ **D)** $5m+6$

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6. **Bitta qisqarmas kasr ko'rinishida yozing:** $\dfrac{x+3}{4}-\dfrac{x-1}{6}$

A) $\dfrac{x+11}{12}$ **B)** $\dfrac{x+7}{12}$

C) $\dfrac{5x+11}{12}$ **D)** $\dfrac{x+4}{10}$

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7. **Bitta qisqarmas kasr ko'rinishida yozing:** $\dfrac{2(3x-1)}{5}-\dfrac{x}{2}$

A) $\dfrac{x-2}{3}$ **B)** $\dfrac{7x-4}{10}$

C) $\dfrac{7x-2}{10}$ **D)** $\dfrac{17x-4}{10}$

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8. $k(2x-5)-4x=2x-15$ **bo'lsa,** k **nechaga teng?**

A) 2 **B)** −3 **C)** 3 **D)** 5

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9. **Qaysi ifoda** $3(2x-1)-(x-5)$ **ga teng kuchli?**

A) $5x-8$ **B)** $5(x+2)$

C) $x+2(2x-3)$ **D)** $2(3x+1)-x$

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10. **Ali shunday yechdi:** $5(2a+3)-4a=10a+3-4a=6a+3$**. Xato qayerda?**

A) 1-qadamda: $10a+8$, javob $6a+8$

B) 1-qadamda: 3 ham 5 ga ko'paytirilishi kerak edi, javob $6a+15$

C) 2-qadamda: javob $14a+3$

D) Xato yo'q

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11. **Velosipedchi 3 soat** $(v+4)$ **km/soat tezlik bilan, keyin 2 soat** $(v-1)$ **km/soat tezlik bilan yurdi. Jami yo'l qaysi ifoda bilan yoziladi?**

A) $5v+3$ **B)** $2v+10$ **C)** $5v+14$ **D)** $5v+10$

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12. **Avval soddalashtiring, so'ng qiymatini toping:** $4(3a-2)-5(2a-3)$, $a=-2{,}5$.

A) -3 **B)** 2 **C)** 12 **D)** -28

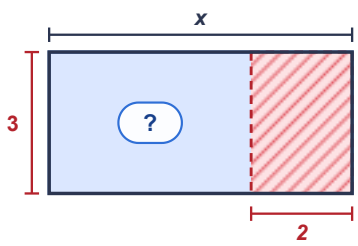
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13. **Teng yonli uchburchakning har bir yon tomoni** $(2x+3)$ **cm, asosi esa yon tomonidan 4 cm qisqa. Uchburchakning perimetri qaysi ifoda?**

A) $5x+5$ **B)** $6x+13$ **C)** $6x+9$ **D)** $6x+5$

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14. **Rasmda eni 3, bo'yi** x **bo'lgan to'g'ri to'rtburchakdan bo'yi 2 bo'lgan qism (shtrixlangan) kesib olindi. Qolgan qismning yuzi qaysi ifoda?**

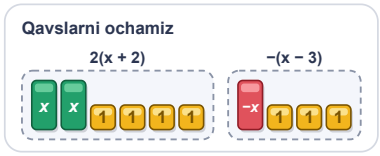


A) $3x+6$ **B)** $x-6$

C) $3x-6$ **D)** $3x-2$

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15. **Rasmda** $2(x+2)-(x-3)$ **ning qavslari tugmalar bilan ochilgan. Nol juftliklarni olib tashlasak, qaysi ifoda qoladi?**



A) $x+1$ **B)** $x-1$

C) $x+7$ **D)** $3x+7$

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16. **Sexda** $(8a+12)$ **metr mato bor edi. Uning** $\frac{3}{4}$ **qismidan ko'ylak, qolganidan sumka tikildi. Ko'ylakka sumkaga qaraganda necha metr ko'p mato ketdi?**

A) $4a+6$ **B)** $4a+3$ **C)** $6a+9$ **D)** $2a+3$

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17. **Bitta qisqarmas kasr ko'rinishida yozing:** $\frac{x-1}{2}+\frac{x+2}{3}-\frac{x-3}{6}$

A) $\frac{x+1}{3}$ **B)** $\frac{4x+4}{11}$

C) $\frac{2x+2}{3}$ **D)** $\frac{2x-1}{3}$

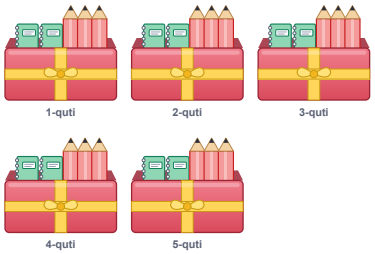
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18. $3(2x-a)+a(x+4)$ **ifoda soddalashtirilganda** x **ning koeffitsiyenti 10 chiqdi. Ozod had nechaga teng?**

A) 16 **B)** -4 **C)** 4 **D)** 28

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19. **Rasmdagi har bir qutida bir xil buyumlar bor. Daftar d so'm, qalam daftardan 1 500 so'm arzon. Hamma qutilar narxi qancha?**



A) $25d - 4\,500$ **B)** $25d + 22\,500$

C) $25d - 22\,500$ **D)** $25d - 7\,500$

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20. **Soddalashtiring:** $\frac{1}{2}[8y - 2(y - 3)]$

A) $3y - 3$ **B)** $3y + 3$ **C)** $6y + 6$ **D)** $4y - 3$

Javoblar

Chiziqli ifodalarda qavslarni ochish · 2-test (qiyinroq)

№	Javob	Qiymati
1	**A**	$-2x+7$
2	**C**	$-a-2b$
3	**B**	$-xy$
4	**C**	$-9x+30$
5	**A**	$7m+6$
6	**A**	$\dfrac{x+11}{12}$
7	**B**	$\dfrac{7x-4}{10}$
8	**C**	3
9	**D**	$2(3x+1)-x$
10	**B**	1-qadamda: 3 ham 5 ga ko'paytirilishi kerak edi, javob $6a+15$

№	Javob	Qiymati
11	**D**	$5v+10$
12	**B**	2
13	**D**	$6x+5$
14	**C**	$3x-6$
15	**C**	$x+7$
16	**A**	$4a+6$
17	**C**	$\dfrac{2x+2}{3}$
18	**C**	4
19	**C**	$25d-22\,500$
20	**B**	$3y+3$

Qisqa yechimlar

1. $-3(2x-5)=-6x+15$ ($(-)\cdot(-)=(+)$), $4(x-2)=4x-8$.
$-6x+4x+15-8$. Javob: $-2x+7$.

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2. $5a-10b-6a+8b$ — ikkinchi qavs oldida -2, ishoralar o'zgaradi.
$5a-6a=-a$, $-10b+8b=-2b$. Javob: $-a-2b$.

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3. $2xy-3x-3xy+3x$.
$2xy-3xy=-xy$, $-3x+3x=0$. Javob: $-xy$.

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4. Avval ichki qavs: $-5(x-2)=-5x+10$. Kvadrat qavsda: $2x-5x+10=-3x+10$.
$3(-3x+10)$ — 3 ni ikkala hadga ko'paytiramiz. Javob: $-9x+30$.

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5. Ichki qavs: $2m-3m-6=-m-6$.
$6m-(-m-6)=6m+m+6$. Javob: $7m+6$.

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6. EKUK(4; 6) = 12: $\dfrac{3(x+3)-2(x-1)}{12}$.
$3x+9-2x+2=x+11$. Javob: $\dfrac{x+11}{12}$.

---

7. EKUK(5; 2) = 10: $\dfrac{4(3x-1)-5x}{10}$.
$12x-4-5x=7x-4$. Javob: $\dfrac{7x-4}{10}$.

---

8. $2kx-5k-4x$: x ning koeffitsiyenti $2k-4=2$, $k=3$.
Tekshiramiz: $-5\cdot 3=-15$. Javob: **3**.

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9. $3(2x-1)-(x-5)=6x-3-x+5=5x+2$.
$2(3x+1)-x=6x+2-x=5x+2$. Javob: $2(3x+1)-x$.

---

10. Qavsdagi **har bir** hadga ko'paytiriladi: $5(2a+3)=10a+15$.
$10a+15-4a=6a+15$. Javob: **1-qadamda xato**.

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11. Yo'l = tezlik · vaqt: $3(v+4)+2(v-1)$.

$3v+12+2v-2=5v+10$. Javob: $5v+10$ km.

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12. $12a-8-10a+15=2a+7$.

$2\cdot(-2{,}5)+7=-5+7$. Javob: **2**.

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13. Asosi: $(2x+3)-4=2x-1$.

$P=2(2x+3)+(2x-1)=4x+6+2x-1=6x+5$. Javob: $6x+5$.

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14. Butun yuz $3x$, kesilgan qism $3\cdot 2=6$.

$3(x-2)=3x-6$. Javob: $3x-6$.

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15. $2x+4-x+3$: bitta x va $-x$ nol juftlik.

Qoladi x va $4+3=7$. Javob: $x+7$.

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16. Ko'ylakka: $\frac{3}{4}(8a+12)=6a+9$. Sumkaga: $8a+12-(6a+9)=2a+3$.

$6a+9-(2a+3)=4a+6$. Javob: $4a+6$ metr.

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17. EKUK = 6: $\frac{3(x-1)+2(x+2)-(x-3)}{6}=\frac{3x-3+2x+4-x+3}{6}$.

$\frac{4x+4}{6}$, 2 ga qisqartiramiz. Javob: $\frac{2x+2}{3}$.

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18. Qavslarni ochamiz: $6x-3a+ax+4a=(6+a)x+a$.

$6+a=10$, $a=4$. Ozod had: $-3a+4a=a=4$. Javob: **4**.

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19. Bitta quti: $2d+3(d-1500)=5d-4500$.

5 ta quti: $5(5d-4500)=25d-22\,500$. Javob: $25d-22\,500$.

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20. Ichki qavs: $8y-2y+6=6y+6$.

$\frac{1}{2}(6y+6)$. Javob: $3y+3$.
